IE3035 Operations Research
HW2

Q1. Show how the following objective function can be presented in equation form:

Minimize $z = \max\{|x_1 - x_2 + 3x_3|, |-x_1 + 3x_2 - x_3|\}$

$x_1, x_2, x_3 \geq 0$

(Hint: $|a| \leq b$ is equivalent to $a \leq b$ and $a \geq -b$.)

Q2. Show that the m equations:

$$\sum_{j=1}^{n} a_{ij} x_j = b_i, i = 1,2,\ldots,m$$

are equivalent to the following $m + 1$ inequalities:

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i, i = 1,2,\ldots,m$$

$$\sum_{j=1}^{n}\left(\sum_{i=1}^{m} a_{ij}\right) x_j \geqslant \sum_{i=1}^{m} b_i$$

Q3. Determine the optimum solution for each of the following LPs by enumerating all the basic solutions.

(a) Maximize $z = 2x_1 - 4x_2 + 5x_3 - 6x_4$

subject to

$x_1 + 4x_2 - 2x_3 + 8x_4 \leq 2$

$-x_1 + 2x_2 + 3x_3 + 4x_4 \leq 1$

$x_1, x_2, x_3, x_4 \geq 0$

(b) Minimize $z = x_1 + 2x_2 - 3x_3 - 2x_4$

subject to

$x_1 + 2x_2 - 3x_3 + x_4 = 4$

$x_1 + 2x_2 + x_3 + 2x_4 = 4$

$x_1, x_2, x_3, x_4 \geq 0$

Q4. Consider the following set of constraints:

$x_1 + 2x_2 + 2x_3 + 4x_4 \leq 40$

$2x_1 - x_2 + x_3 + 2x_4 \leq 8$

$4x_1 - 2x_2 + x_3 - x_4 \leq 10$

$x_1, x_2, x_3, x_4 \geq 0$

Solve the problem for each of the following objective functions.

(a) Maximize $z = 2x_1 + x_2 - 3x_3 + 5x_4$

(b) Maximize $z = 8x_1 + 6x_2 + 3x_3 - 2x_4$

(c) Maximize $z = 3x_1 - x_2 + 3x_3 + 4x_4$

(d) Minimize $z = 5x_1 - 4x_2 + 6x_3 - 8x_4$

Q5. The following tableau represents a specific simplex iteration. All variables are nonnegative. The tableau is not optimal for either a maximization or a minimization problem. Thus, when a nonbasic variable enters the solution it can either increase or decrease z or leave it unchanged, depending on the parameters of the entering nonbasic variable.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	Solution
z	0	-5	0	4	-1	-10	0	0	620
x_8	0	3	0	-2	-3	-1	5	1	12
x_3	0	1	1	3	1	0	3	0	6
x_1	1	-1	0	0	6	-4	0	0	0

(a) Categorize the variables as basic and nonbasic and provide the current values of all the variables.

(b) Assuming that the problem is of the maximization type, identify the nonbasic variables that have the potential to improve the value of z. If each such variable enters the basic solution, determine the associated leaving variable, if any, and the associated change in z. Do not use Gauss-Jordan row operations.

(c) Repeat part (b) assuming that the problem is of the minimization type.

(d) Which nonbasic variable(s) will not cause a change in the value of z when selected to enter the solution?